

FIRST SWING AT OBLIGATION O1: IS THE LARGE- β REDUCTION OF THE TW_β TAIL TO PAINLEVÉ II *ALL-ORDERS* EXACT?

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ABSTRACT. Working note toward Theorem T1 of the TW_β skeleton. O1 asks whether the Comtet–Le Doussal–Smith (CLDS) weak-noise reduction — which they establish for the leading rate function $\Phi(a)$ (governed by Painlevé II) and the $O(1/\beta)$ cumulants — extends to *all orders* in $\varepsilon^2 = 1/\beta$, so that Costin’s rank-one ODE Borel-summability applies. This note (i) writes the exact reduction of the tail to a backward-Kolmogorov ODE for the Riccati diffusion; (ii) carries out the leading weak-noise WKB, recovering the instanton as an *Airy* log-derivative and the leading rate $\Phi(a) = \frac{2}{3}a^{3/2}$ (matching Dumaz–Virág); (iii) sets up the all-orders transport hierarchy and shows its coefficients are iterated Airy integrals; and (iv) reduces O1 to one sharp, self-contained sub-problem — exhibit the single rank-one nonlinear ODE whose trans-series coefficients are the fluctuation coefficients $c_n(a)$ — which is where the genuine work remains. Verdict: leading order and the generating structure are in hand; the “single ODE” identification is the open crux, with a concrete plan to close it.

1. THE EXACT REDUCTION (NO APPROXIMATION YET)

Let $\Lambda_0(\beta)$ be the lowest eigenvalue of $\mathcal{A}_\beta = -\partial_x^2 + x + 2\varepsilon b'(x)$ on $L^2(\mathbb{R}_+)$, Dirichlet at 0, $\varepsilon = \beta^{-1/2}$. Fix a level $\Lambda = -a$ and introduce the Riccati variable $p = \psi'/\psi$ for $-\psi'' + (x + 2\varepsilon b')\psi = \Lambda\psi$. Then, as an Itô SDE in the “time” x ,

$$dp = ((x + a) - p^2) dx + 2\varepsilon dB_x, \quad p(0^+) = +\infty, \quad (1)$$

where B is the Brownian motion of b' . The Sturm oscillation/Riccati dictionary gives the *exact* identity

$$\mathcal{T}_\beta(a) := \mathbb{P}(\Lambda_0(\beta) < -a) = \mathbb{P}(\text{1 explodes to } -\infty \text{ on } [0, \infty)). \quad (2)$$

Writing $u(p, x) := \mathbb{P}(\text{no explosion} \mid p_x = p)$, the backward Kolmogorov equation is the *exact* linear PDE

$$\partial_x u + ((x + a) - p^2) \partial_p u + 2\varepsilon^2 \partial_p^2 u = 0, \quad (3)$$

with $u \rightarrow 1$ away from the explosion boundary and $u \rightarrow 0$ as $p \rightarrow -\infty$; $\mathcal{T}_\beta(a) = 1 - u(+\infty, 0)$. Everything below is the $\varepsilon \rightarrow 0$ (large- β) analysis of (3). [established here]

2. LEADING WEAK-NOISE WKB: THE INSTANTON IS AIRY, THE RATE IS $\frac{2}{3}a^{3/2}$

Insert the WKB ansatz $u = \exp(-W(p, x)/\varepsilon^2)(A_0 + \varepsilon^2 A_1 + \varepsilon^4 A_2 + \dots)$ into (3). Order ε^{-2} gives the eikonal (Hamilton–Jacobi) equation

$$\partial_x W + ((x + a) - p^2) \partial_p W + 2(\partial_p W)^2 = 0, \quad (4)$$

i.e. $-\partial_x W = H(p, \partial_p W, x)$ with the Freidlin–Wentzell Hamiltonian $H(p, \theta, x) = \theta b + 2\theta^2$, $b := (x + a) - p^2$. Zero-energy trajectories $H = 0$ split into the relaxation branch $\theta = 0$ ($\dot{p} = b$) and the *fluctuation* branch $\theta = -b/2$, on which

$$\dot{p} = \partial_\theta H|_{\theta=-b/2} = b + 4\theta = -b = p^2 - (x + a). \quad (5)$$

Equation (5) is a Riccati equation; the substitution $p = -\varphi'/\varphi$ linearizes it to the **Airy equation**

$$\varphi'' = (x + a)\varphi, \quad (6)$$

so the instanton is $p_\star(x) = -\varphi'/\varphi$ for the Airy combination that carries p from the quasi-stable branch $p_+ = +\sqrt{x+a}$ down to explosion ($p \rightarrow -\infty$). [established here]

Lemma 1 (Leading rate). $W_0(a) := W(+\infty, 0) = \frac{1}{2} \int_0^\infty b(x)^2 dx$ along (5) equals $\frac{2}{3}a^{3/2}$ to leading order in a .

Derivation. On the fluctuation branch $dW = \theta dp$ (since $H = 0$), and $\theta = -b/2$, $\dot{p} = -b$, so $W_0 = \int \theta \dot{p} dx = \frac{1}{2} \int b^2 dx$. Frozen at the boundary $x = 0$, (1) is a gradient escape $dp = -U'(p) dx + \sqrt{2D} dB$ with $U(p) = \frac{1}{3}p^3 - ap$, $D = 2$ (after scaling out ε^2), and escape action $\Delta U/D$. The barrier is $\Delta U = U(-\sqrt{a}) - U(+\sqrt{a}) = \frac{4}{3}a^{3/2}$, whence the rate is $\frac{4}{3}a^{3/2}/2 = \frac{2}{3}a^{3/2}$. The boundary crossing dominates because the barrier $\frac{4}{3}(x+a)^{3/2}$ is smallest at $x = 0$. $\square$

This matches the rigorous Dumaz–Virág exponent $\mathcal{T}_\beta(a) = a^{-3\beta/4+o(1)}e^{-\frac{2}{3}\beta a^{3/2}}$ [1]: the eikonal reproduces the leading exponential, and the $a^{-3\beta/4}$ prefactor is the one-loop factor A_0 of §3. [known, cited] / [established here]

Remark 2 (Where CLDS’s Painlevé II sits). CLDS [2] show the full rate function $\Phi(a)$ — as a function of the *level* a , interpolating bulk to tail — solves Painlevé II; $\frac{2}{3}a^{3/2}$ is its $a \rightarrow \infty$ asymptotic (the Hastings–McLeod tail). So PII is the ODE in the *spectral parameter* a , not the instanton ODE in x (which is Airy, (6)). This is the crucial structural point for O1: the a -dependence, not the x -profile, carries the integrable backbone. CLDS compute Φ and the $O(1/\beta)$ cumulants only; orders ε^4 and beyond are untouched. [known, cited]

3. THE ALL-ORDERS TRANSPORT HIERARCHY

Collecting orders $\varepsilon^0, \varepsilon^2, \dots$ in (3) along the instanton characteristics $X \mapsto (p_\star(X), X)$ gives a hierarchy for the A_n . Writing $\mathcal{L}_0$ for the first-order transport (Liouville) operator $\partial_x + (b - 4\theta_\star)\partial_p$ along the instanton and $\Delta := \partial_p^2$:

$$\mathcal{L}_0 A_0 = \kappa A_0, \quad \mathcal{L}_0 A_n = \kappa A_n - 2\Delta A_{n-1} \quad (n \geq 1), \quad (7)$$

where $\kappa = -2\partial_p^2 W$ is the local expansion rate of the flow (the one-loop density). Thus:

- A_0 is the **one-loop fluctuation determinant** of the instanton — the Van Vleck/Gelfand–Yaglom factor of the Jacobi operator $\mathcal{J} = -2\partial_p^2 + (\text{Hessian of } H \text{ along } p_\star)$. Its zero mode (instanton translation) is removed within the Riccati formulation as in Schorlepp–Grafe–Grauer [6] (obligation O4), leaving the finite $a^{-3\beta/4}$ prefactor. [known, cited]
- each A_n ($n \geq 1$) solves a *linear* transport ODE (7) driven by A_{n-1} ; since $\mathcal{L}_0$, κ , and Δ are all built from the instanton $p_\star = -\varphi'/\varphi$ with φ Airy (6), and the transport Green’s function is the Airy Green’s function, **each** $c_n(a) = A_n(+\infty, 0)$ **is an iterated integral of Airy functions over the instanton**. [established here]

Claim 3 (Structural all-orders statement). *The fluctuation coefficients $\{c_n(a)\}_{n \geq 0}$ are generated, to all orders, by the transport hierarchy (7) along the Airy instanton (6); each is a finite iterated Airy integral (after O4 zero-mode removal). Hence the tail admits the all-orders WKB form $\mathcal{T}_\beta(a) = e^{-\beta\Phi(a)} \sum_{n \geq 0} c_n(a) \beta^{-n}$ with computable c_n . [established here] (modulo O4)*

4. WHAT O1 STILL NEEDS: ONE SHARP SUB-PROBLEM

Claim 3 gives an all-orders *recursion*, but Costin’s Borel-summability theorem [4] is a statement about the trans-series of a *single rank-one nonlinear ODE at an irregular singularity*, not about a linear transport hierarchy. The remaining content of O1 is exactly the bridge between the two:

Sub-problem O1*. *Exhibit a single nonlinear ODE, in the spectral variable a , of rank-one/irregular-singular type, whose formal trans-series coefficients coincide with the $c_n(a)$ of Claim 3. The expected ODE is the Painlevé II of Remark 2; the expected irregular singularity is $a \rightarrow \infty$.*

Two independent reasons this is the right target, and not wishful:

- (R1) CLDS already prove the *leading* datum $\Phi(a)$ solves PII; the transport hierarchy (7) is a linearization about that solution, and the linearization of a Painlevé equation about a fixed solution is governed by the *same* isomonodromy data — so the c_n should be the PII trans-series coefficients, not merely Airy integrals that accidentally resum. [GAP] (to verify)
- (R2) The deterministic Hastings–McLeod/PII trans-series is *proved* to have exactly the reality structure our c_n must have — factorially divergent, non-sign-alternating in the $a \rightarrow \infty$ region, with the median-resummation reality condition (Cleri–Dunne [3]). A linear hierarchy has no reason to reproduce that nonlinear Stokes structure; a PII identification does. [known, cited]

Once O1* holds, Costin [4] gives Borel summability of the c_n -series in $1/\beta$ with singularities at the instanton action, Costin–Costin–Kohut [5] pins the Stokes constant $\neq 0$, and Theorem T1 follows.

Concrete route to O1* (three checkable steps).

Step 4 (Isomonodromy identification). Write the second variation of the FW action (the Jacobi operator $\mathcal{J}$ of §3) in the Lax/isomonodromy frame of the PII solution $\Phi(a)$. If $\mathcal{J}$ is the linearized-PII operator about Φ , then (7) is the PII trans-series recursion. This is a finite computation: compare the coefficients of $\mathcal{J}$ (built from $p_* = -\varphi'/\varphi$) against the PII linearization. [GAP]

Step 5 (Rank-one / irregular-type check). Verify that the resulting ODE in a has a rank-one irregular singularity at $a = \infty$ (Painlevé transcendents do), so Costin’s hypotheses [4] are literally met. [GAP]

Step 6 (Cross-check against the numerics). The companion cusp engine can compute $c_1(a), c_2(a)$ for the Airy ($q=1$) case directly (the same Wiener-chaos machinery, one dimension down); check they match the PII trans-series coefficients predicted by Steps 1–2. This is a decisive falsifiable test before committing to the proof. [established here] (tooling exists)

5. VERDICT

Established here: the exact reduction (3); the leading instanton (Airy, (6)) and rate $\frac{2}{3}a^{3/2}$ (Lemma 1, matching [1]); the all-orders transport hierarchy with c_n as iterated Airy integrals (Claim 3). *The crux that remains (O1*):* identifying the single rank-one nonlinear ODE — almost certainly the PII of [2] — whose trans-series is $\{c_n\}$, reducing O1 to the finite isomonodromy computation of Step 1 plus the numerical cross-check of Step 3. *Assessment:* O1 is **not** a multi-year problem; it is a bounded computation gated on Step 1, and Step 3 gives an immediate go/no-go using existing tooling. Recommend doing Step 3 (numerical c_1, c_2 vs. PII prediction) first as a cheap falsifier, then Step 1 as the proof.

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